

Duc-Anh Nguyen

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RESEARCH INTERESTS

My research focuses on uncovering the mathematical principles underlying foundation models, leveraging these insights to design efficient, interpretable, and robust LLMs and VLMs, with an emphasis on optimization, sparse modular architectures, and transfer learning.

EDUCATION

- **University of Virginia, Charlottesville (UVA)** Aug 2026
Ph.D. in Computer Science Virginia, USA
 - Advisor: [Prof. Hadi Daneshmand](#)
- **Hanoi University of Science and Technology (HUST)** Oct 2020 – Sep 2024
B.S. in Mathematics and Informatics (Talented Program) Hanoi, Vietnam
 - **Valedictorian** (Ranked 1st), Cumulative GPA: 3.92/4.00.
 - Scholarships for students with outstanding academic achievement every semester.
 - Thesis: Steepest Descent for Multi-Objective Optimization and Machine Learning Applications (**Best Presentation Award**).

RESEARCH EXPERIENCE

- **AI Research Resident** Mar 2024 – Present
VinAI Research (acquired by Qualcomm) Hanoi, Vietnam
 - Supervisor: [Dr. Toan Tran](#).
 - Research Topic: Efficiency, Interpretability, and Robustness of Foundation Models (Sparse Mixture of Experts, Transformers and State-Space Models).
- **Research Student** Aug 2022 - Aug 2024
Modeling, Simulation and Optimization in Data Science (HATLAB), HUST Hanoi, Vietnam
 - Supervisor: [Dr. Thang Tran Ngoc](#).
 - Research Topic: Multi-objective Optimization in Machine Learning.

PUBLICATIONS & PATENTS (ORCID: [0009-0009-0528-8430](#))

(*) denotes equal contribution

1. **Mixture-of-Control: State-Aware Fine-Tuning for Transformer-based Models**
Duc-Anh Nguyen*, Tien-Ngoc Luu*, Tung Pham, Toan Tran
Approved for Filing Patent, Qualcomm, 2026
2. **Artificial Intelligence in Metal Additive Manufacturing: Current Status, Challenges, and Future Developments**
Duc-Anh Nguyen*; Bui Truong Giang Le*; Van Anh Nguyen; Minh Tuan Vu; Manh Ha Bui; Minh Thanh Le; Trong Duc Nguyen; Tien Dung Hoang; Xuan Hai Le.
Journal of Intelligent Manufacturing (Q1), Impact Factor: 7.4.
3. **Selective Sinkhorn Routing for Improved Sparse Mixture of Experts**
Duc-Anh Nguyen*, Huu Binh Ta*, Nhuan Le Duc, Tan Minh Nguyen, Toan Tran
Approved for Filing Patent, Qualcomm, 2025

4. **Selected Beautiful Math Problems from International Competitions (Geometry)** [Book]
 Nguyen Duy Khuong, Phan Quang Tri, Tran Quoc Dung, Nguyen Duong Minh, Pham Nguyen Phuc Long, **Duc-Anh Nguyen**
 Vietnam National University Press, 2025
5. **Determinants of Credit Risk under the Basel II Accord: A Case Study of the Vietnamese Banking Sector**
 Ngo Thu Giang, **Duc-Anh Nguyen**, Vu Thi Thao Chi, Nguyen Bao Anh, Nguyen Tai Quang Dinh
 International Conference on Emerging Challenges in Economics and Business (ICECH 2023)
6. **Adaptive Conflict-Averse Multi-gradient Descent for Multi-objective Learning**
 Dinh Van Tuan, Tran Anh Tuan, **Duc-Anh Nguyen**, Bui Khuong Duy, Tran Ngoc Thang
 International Conference on Intelligence of Things (ICIT 2023)
7. **Exploration of Plane Geometry** [Book]
 Le Xuan Hoang, **Duc-Anh Nguyen**
 Vietnam National University Press, 2022

PREPRINTS

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| Mixture-of-Control: State-Aware Fine-Tuning for Transformer-based Models
Duc-Anh Nguyen* , Tien Ngoc Luu*, Tung Pham, Toan Tran | 2026 |
| Selective Sinkhorn Routing for Improved Sparse Mixture of Experts
Duc-Anh Nguyen* , Huu Binh Ta*, Nhuan Le Duc, Tan Minh Nguyen, Toan Tran | 2025 |
| Incremental attribute reduction on dynamic decision tables with intuitionistic fuzzy weighted neighborhood rough sets
Viet Anh Pham, Ngoc Thuy Nguyen, Duc-Anh Nguyen , Long Giang Nguyen | 2025 |
| Artificial Intelligence for Metal Material Science: Challenges, Solutions, and Future Developments
Bui Truong Giang Le*, Duc-Anh Nguyen* , Minh Tuan Vu, Phan Anh Tran, Trong Duc Nguyen, Minh Thanh Le, Khue Nguyen Vo Ngoc, Duy Hung Tran, Van Anh Nguyen | 2025 |
| A Steepest Gradient Method with Nonmonotone Adaptive Step-sizes for the Nonconvex Minimax and Multi-Objective Optimization Problems
Duc-Anh Nguyen , Tran Ngoc Thang | 2025 |

TALKS

- Workshop on Scientific Computing and Applications (Talk in session with discussion): “*The Minimax Approach for solving non-convex multiobjective optimization and Applications*”, 2024
- VIASM Summer school “Research Experience for Undergraduates” (Talk in session with discussion): “*The Steepest Descent Method for Solving Nonconvex Minimax Problem and Applications to Multi-Task Learning*”, 2023

HONORS & AWARDS

- **National Program for the Development of Mathematics Scholarship** Nov 2023 – Apr 2024
 Vietnam Institute for Advanced Study in Mathematics (VIASM)
 - Awarded twice for outstanding academic performance in 2023–2024.
- **Gold Medal - International Youth Math Challenge** Nov 2023
 - Final-round participant; ranked among top 2% of all participants.
- **Exness Scholarship** Aug 2023
 Hanoi University of Science and Technology & Exness Corporation

- **Third Prize - VIASM Summer School “Research Experiences for Undergraduates”** Aug 2023
Vietnam Institute for Advanced Study in Mathematics (VIASM)
- **Gold Medal - Iranian Combinatorics Olympiad** Jun 2022

LANGUAGES

- Vietnamese: Native.
- English: IELTS Academic **7.0 Overall** (Listening 7.0, Reading 7.0, Writing 6.5, Speaking 6.5).

LEADERSHIP & INVOLVEMENT

- **Member** Nov 2022 – Dec 2024
FPT Young Talents Hanoi, Vietnam
 - Selected for a highly competitive 2-year talent development program within FPT, gaining exposure to leading researchers and expanding professional networks.
 - Equipped with a growth mindset through mentorship and participated in multiple FPT hackathons, strengthening coding and mathematical modeling skills.
- **Co-organizer** Mar 2023 – Aug 2023
Hanoi Mathematical Modeling Hanoi, Vietnam
 - Led the organizational development efforts, including preparing and writing academic content, presenting at internal seminars on mathematical modeling.
 - Contributed to operational planning and supported participants throughout testing phases.

PROFESSIONAL SERVICES

Reviewer, CoLoRAI Workshop at ICML 2026

REFERENCES

- **Dr. Toan Tran**
Senior Staff Engineer, Qualcomm AI Research
toatran@qti.qualcomm.com
- **Dr. Tung Pham**
Research Scientist, Qualcomm AI Research
tungp@qti.qualcomm.com
- **Dr. Thang Tran Ngoc**
Lecturer, Faculty of Mathematics & Informatics
Hanoi University of Science and Technology
thang.tranngoc@hust.edu.vn
- **Prof. Van Anh Nguyen**
Lecturer, University of Kentucky, USA
vananh.nguyen@uky.edu